بإمكانكم الأطلاع على تسجيلات المادة
من خلال منصة جو اكاديمي

CHEMISTRY 101

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What is the wavelength of the emitted light produced when an electron moves from n = 3 to n = 1 in hydrogen atom:

$$\frac{1}{\lambda} = \mathbf{1.097 * 10^7 \left(\frac{1}{ni^2} - \frac{1}{nf^2}\right)}$$

$$\frac{1}{\lambda} = \mathbf{1.097 * 10^7 \left(\frac{1}{9} - \frac{1}{1}\right)} ==== (\mathbf{-9751111.1})$$

$$\frac{1}{\lambda} = \mathbf{9653600} \ldots\ldots\ldots\ldots \ \boldsymbol{\lambda} = \mathbf{1.0255 * 10^{-7}}$$

Q2: A sample is composed of (38.7 % C), (9.7 % H) and (51.6 % O) by mass, its molar mass is 62.1 g /mole, what is the empirical formula of this sample?

	C	H	O
%	38.7	9.7	51.6
Mass	38.7 g	9.7 g	51.6 g
Mole	38.7/12 = 3.225 mole	9.7 / 1 = 9.7 mole	51.6 / 16 = 3.225 mole
اقل عدد مولات هو (3.225)			
# of atoms	3.225 / 3.225 = 1	9.7 / 3.225 = 3	3.225 / 3.255 = 1
CH_3O			

- Molar Mass of CH3O = 31 g/mole

  Ratio of molar mass = (62.1 / 31) = 2

Compound = 2 * (CH_3O) = $C_2H_6O_2$

30 ml of 0.12 M HCl solution are required to neutralize 36 ml of NaOH solution, what is the molarity of NaOH solution?

M1 = 0.12   M2 =

V1 = 30 ml   V2 = 36 ml

$$M1 * V1 = M2 * V2$$

$0.12 * 30 = M2 * 36\,ml \ldots\ldots\ldots\ldots\ldots\ldots M2 = 0.1\,M$

Calculate the velocity of moving neutron ($1.67 * 10^{-27}$ Kg) whose wavelength is 500 pm:

$\boldsymbol{\lambda}$ = 500 pm = 500 $*10^{-12}$ m

$$\lambda = \frac{h}{mv} ===> v = \frac{h}{m\boldsymbol{\lambda}}$$

$$v = \frac{6.626 * 10^{-34}\,J * S}{1.67 * 10^{-27} Kg * 500 * 10^{-12} m} * \frac{Kg.m^2}{S^2} = 793.5\frac{m}{s}$$

How many unpaired electrons are there in ground state electronic configurations of phosphors atom?

P (15 electrons) = $1S^2\ 2S^2\ 2P^6\ 3S^2\ 3P^3$

# of unpaired electrons = 3

Which one of the following elements is metalloid?

Cs Ge W Hg

Answer = Ge

How many grams of H_2O must be consumed in order to produce 6.5 g C_2H_2 according to this equation?

$$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$$

- Molar Mass of C_2H_2 = 26 g /mole
- Mole of C_2H_2 = Mass / molar Mass = 6.5 / 26 = 0.25 mole
- 2 mole $H_2O \rightarrow$ 1 mole C_2H_2

  ?? Mole $H_2O \rightarrow$ 0.25 mole C_2H_2
- # of mole H_2O = 0.5 mole
- Mass of H_2O = 0.5 * 18 = 9 gram

The theoretical yield In g of (NH_3) from the reaction of 7.1 g (N_2) and 36 g (H_2) according this equation is ($N_2 + 3\ H_2 \rightarrow 2\ NH_3$)

- Molar Mass of N2 = 28 g/mole
- Molar Mass of H2 = 2 g/mole

- Mole of N2 = 7.1 / 28 = 0.254 mole

- Mole of H2 = 36 / 2 = 18 mole

- Limiting reactant = N2

- 1 Mole N2 → 2 mole NH3

  0.254 mole N2 →?? Mole NH3

  Mole of NH3 = 0.508 mole

  Mass of NH3 = Theoretical yield = 0.508 * 17 = 8.636 g

A certain liquid has density of 2.67 g/cm^3, 30.5 L of this liquid would have a mass ……?

1 L = 1000 Cm ^{3}

$$Density = \frac{Mass}{Volume} ===> Mass = Density * Volume$$

$$2.67 \frac{g}{cm^3} * 30.5\ L * \frac{1000\ cm^3}{1L} = 81435\ g = 81.435\ Kg$$

How many significant figures are in (0.00340501)?

6 significant figures

750 m/s = ………. Km/hour

$$\frac{750\ m}{s} * \frac{3600\ s}{1 hour} * \frac{1\ Km}{1000\ m} = 2700\,\frac{Km}{hour}$$

The energy of 1 mole of photons that has a wavelength of 0.02 m is ……… J

$$E = h * \left(\frac{C}{\boldsymbol{\lambda}}\right) * \mathbf{6.02} * \mathbf{10^{23}}$$

$$E = 6.626 * 10^{-34} * \left(\frac{3 * 10^{8}}{0.02}\right) * 6.02 * 10^{23}$$

E = 5.983 J